

T6: Making the most of cis-regulatory annotations for genome-scale analysis

A practical day for moving from epigenomic evidence to reusable regulatory annotation workflows.

ECCB 2026 Tutorial

03/09/2026

Tutorial Instructors



Garth Ilesley

Project Leader, Ensembl Regulation, EMBL-EBI

- Regulatory genome annotation at scale
- Enhancer biology, reproducible analysis and community resources



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Senior Bioinformatician, Ensembl Regulation, EMBL-EBI

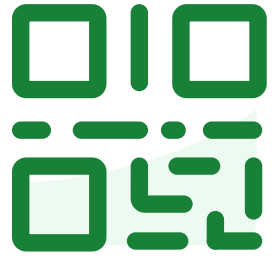
- Regulatory annotation pipelines and quality assessment
- R, statistics, data integration, and training



Lousse Paola Mirabueno

Ensembl Outreach Officer, Ensembl Outreach, EMBL-EBI

- Ensembl training development and outreach
- Scientific communication and creation of documentation, tutorials, and interactive learning materials



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#1350613**



What do you hope to gain from today's workshop?

Tutorial Goals

- Regulatory annotations are powerful because they compress messy evidence into usable genomic features.
- The challenge is not access. The challenge is interpretation:
 - Multiple repositories
 - Different annotation frameworks
 - Cell-type and species context
 - Files, browsers, and programmatic workflows

Tutorial Goals

- By the end of the day you should leave with three practical moves

	FIND Explore raw and processed epigenomic resources without getting lost in repository details	• Ensembl Genome Browser • Ensembl Regulation subsite • ENCODE and FAANG Data Portal • ENCODE SCREEN
	INTERPRET Understand what the annotation is saying, which evidence supports it, and where limitations remain.	• Regulatory features vs supporting evidence • cCREs and classification schemes • Cell-type and species context
	REUSE Bring annotations into your own genomic regions, variants, and browser sessions.	• R/GenomicRanges • Python/Pandas/PyRanges • IGV and user-defined regions

Agenda

Coffee break

Lunch break

Coffee break

09:00

09:15

10:00

10:30

10:45

11:30

12:45

13:45

14:45

15:15

15:30

17:15

Welcome

Regulatory genomics evidence

Assays, signals and peaks, CREs inference

ENCODE: Experimental Data, cCREs, and SCREEN

The ENCODE Data Portal. Available assays, metadata and methods. Reference epigenomes

cCREs annotation and the SCREEN portal.

Ensembl Regulation: Regulatory Builds and Evidence

The Ensembl Project. Ensembl Regulation pipelines. Regulatory data, activity and features. The Ensembl Regulation subsite. Comparison of regulatory annotation frameworks.

Alternative Access to Regulatory Annotations

CREs repositories and files. File formats. Data analysis in Python/R.

Visualization with Genome Browsers

IGV for integrating epigenomics evidence and CREs annotation. Customisation alternatives and interpretation

Case Study: Non-coding Variant Interpretation

Conceptual interpretation, resources navigation and utilization, file analysis workflows and genome visualization.

Wrap-up

Tutorial repository

[ECCB-2026-T6 GitLab repository](#)

— data/	Tutorial datasets
— docs/	Installation guides and additional documentation
— hands-on-solutions/	Presentations with hands-on exercises solutions
— notebooks/	Google Colab notebooks
— scripts/	R, Python and shell scripts
— slides/	Presentation slides
— tutorial_information/	Tutorial programme and schedule

Setup checks

- ✓ Internet access
- ✓ Web browsers:
 - Ensembl
 - Ensembl Regulation
 - ENCODE SCREEN
- ✓ IGV
- ✓ Google Account - Google Colab
- ✓ Access to the tutorial material

Questions?

